Problem set 7 (due May 9)

1. (30pts) Consider a spherically symmetric planet of radius R and uniform mass density ρ.

   (a) (20pts) Find the potential energy of a particle of mass m at a position $\vec{r}$ in the interior of the planet.

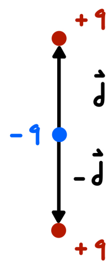
   (b) (10pts) Compute the force associated with this potential. The force should agree with the expression computed in class using Gauss' law.

2. (20pts) Consider a uniform, i.e. constant, electric field $\vec{E}$.

   (a) (10pts) Find the potential energy of a particle with charge q in this electric field.

   (b) (10pts) Using the expression above, find the potential energy of a dipole with moment $\vec{p}$ in this electric field.

3. (50pts) Consider the configuration of charges shown in the diagram below where q is a positive number.



   (a) (10pts) Sketch the electric field lines.

   (b) (10pts) Write the electric field $\vec{E}$ as a function of the radial coordinate $\vec{r}$ (make sure to indicate the origin of your coordinate system).

   (c) (15pts) The electric field $\vec{E}$ at a distance $r \gg |\vec{d}|$ can be expanded in powers of $1/r$. The leading term in the expansion falls off as $1/r^n$ for some integer n. Find the leading term. Explain why we should expect this result.

   (d) (15pts) Compute the next term in the expansion of the electric field at a distance $r \gg |\vec{d}|$, that is, the term that goes like $1/r^{n+1}$, where n was determined above. Explain why we should expect this result.